ACTION A/V: AN IMPROVISATORY PROCESS FOR TRANSLATING MOVEMENT TO AN AUDIOVISUAL OUTCOME

Roger Alsop
School of Performing Arts
Victorian College of the Arts and Music
University of Melbourne

ABSTRACT

This paper discusses the computer based audio-visual improvising system *Action A/V* and the process of developing a series of improvised 'action audio-visual' works. It introduces the conceptual background to the process, discusses precedents and how *Action A/V* is differentiated form those precedents, and outlines how the system was made, and some testing of the system.

Action A/V is a process that builds on the idea of action painting, translating dynamic and ephemeral actions into an audio-visual event. It was designed to allow the processes of developing an artwork in one mode to create an artwork in a different mode.

In early tests and experiments, carried out at RMIT and discussed here, the movements of participants from a dance background generated the actions that were translated and then stored in a movie format.

The process through which these audio-visual events are triggered is designed to have similarities to the process of collaborative improvisation, in that the sounds and screen the improviser are interacting with are to some extent predicable and learnable by the participant. This allows participants to improvise with sonic and visual events in much the same way as they might improvise with another participant, creating a complementary interaction between themselves and the computer-based output.

1. INTRODUCTION

The term '*Action A/V'* harks from the process of 'action painting', made famous by Jackson Pollock, as shown in Hans Namuth's film *Jackson Pollock* [8]. In this film the direct relationship between Pollock's movement and the simultaneous creation of an image is seen to great effect.

Namuth's film, and the many photographs he took of Pollock in the act of painting, show the great mastery Pollock had of his technique and his ability to exercise it expressively [6]. This technical mastery allowed a unique expression to occur, one that caused a re-consideration of a painting from being a finished, representative, work to one that depicted the artist's process as being a major part of the work; a process where the canvas becomes "a space in which to … "express" an object, actual or imagined. What was to go on the canvas was not a picture but an event." [12]

This 'abstract expressionism', a term that followed on from and subsumed the term 'action painting' and is often ascribed to Pollock's work, was also ascribed to dancer/choreographer Martha Graham, a contemporary of Pollock [10]. Graham's work was intensely introspective, being based on, expressing, and being mediated by, the author's often-immediate internal, personal, and subjective, experience. In the case of Pollock and Graham this expression could not be given voice through literal forms or reification, instead they took an abstract form, requiring and allowing shape and movement to be the expressive tools of their intentions.

The obvious differences between Pollock's and Graham's work is that one is concrete and static, and the other ephemeral and dynamic. However, when viewing Namuth's film it is possible to see Pollock's movement in a similar way to that in which one might view a talented dancer; his gestures demonstrating great control, focus and intention, similar to that of a quality dancer. When viewing Pollock's movements it can be seen that the canvas becomes "an arena in [or on] which to act" [12], much like space is the arena in which a dancer acts. The canvas takes and documents Pollock's dynamic actions, translating them into colours and shapes on the canvas, therefore making them static and concrete. 'Action A/V' expands on this concept and process, creating a dynamic, temporal, but not ephemeral, audio-visual work, through concretizing the ephemeral actions of a participant.

2. PRECEDENTS TO *ACTIONA/V* AND ITS PROCESS

A great deal of work has been done in the area of mapping action to sound and visuals, the plethora of books, web pages, and journal articles dedicated to the topic attests to this. A comprehensive listing of writings on the topic up to 2004 can be found at *Interactive Systems and Instrument Design* in the sub heading *Dance Technology* [21]; a listing of practitioners and software can be found at *Dance Technology* [19]. While this site's focus is in the area of music, discussions include haptics, neurology, meaning (and subsequent interpretation) and facilitation through virtual computer based, and actual, instrument/interface making.

Local activity in the relationship between movement/dance and computer based interactive

technologies can be seen locally in the work of Company in Space, Hellen Sky, Garth Paine, Critical Path, Chunky Move, autraLYSIS, and many more. And international activity includes the work of Troika Ranch, Nick Rothwell, Sarah Rubidge, and Johannes Birringer, and again, many more. Clearly these lists leave out too many very worthy practitioners, theoreticians, and academics. However, each of these artists, and many who have not been listed, contributes unique and idiosyncratic approaches, from both conceptual and technical points of view.

2.1.1. Two examples of the relationship between computers, action, and sound

There are a wide variety of roles that these practitioners take, and often the technology providers take a collaborative position, taking the role of enhancing the performance through their technological skills and the conceptual frames, paradigms, and processes that they bring to whatever project they are involved in.

Two examples of the relationships between artists engaged in computer based technologies collaborating with artists engaged in the more traditional areas of performance arts are: Nick Rothwell's work on *the Quartet Project*, and the relationships between Frieder Weiss and the dance company Chunky Move in the work *Mortal Engine*.

In *the Quartet Project* Nick Rothwell [13] worked with the Australian based media designer Margie Medlin, director of the work, and other sound, music, dance and media artists and programmers to produce a work "Informed by new scientific research in the field of physiology, this multi-media event is the culmination of several years collaborative research. [It investigated] the kineasthetics of music: determining movements which produce sounds, which in turn produce new choreographies." The project incorporated "Specialists from dance, music, biomedical and computational science, 3D animation and motion control" [13] .

When discussing the project, Rothwell [1, 14] outlined the way in which each of the four elements in the quartet influenced and interacted with each other, forming feedback between the different elements and performers, and potentially forming a complementary forward feed through the influence of the technology on the potential decisions of the musicians, dancers and virtual/robotic elements in the work.

The performance work *Mortal Engine* has two distinct sections, one, which is primarily oriented towards dance and projected images, followed by the other, which is oriented towards sound/music and visuals. In this piece, as in its predecessor Glow, choreographer and artistic director of Chunky Move Gideon Obarzanek, considered using video as a substitute for standard theatre lighting, and developed this idea to the point where the video became as much a part of, and agent in, the choreography as the movements of the dancer.

It was described as having "no pre-rendered video, light or laser images and the music is similarly generated by movement. Pre-composed phrases are triggered by the dancers' motions, or by the operator at the correct point in the performance. " [5]. Even so there is a great deal of interaction between the dancers and the other elements that is prescribed and rehearsed, with all events following a score, or procedure, that includes the interactive programming of Frieder Weiss.

Eyecon (© Freider Weiss) was the programme used in *Mortal Engine*, its "main use has been to facilitate interactive performances and installations in which the motion of human bodies is used to trigger or control various other media (music, sounds, photos, films, lighting changes, etc.)" [22].

In both cases the software designers developed systems that were both idiosyncratic to the designer or individualized to the project. Rothwell's approach was to design a system specific to and reacting to the outcomes required by the project. This was done in close collaboration with the director and the other participants, developing the artistic work and the tools with which to make it simultaneously.

Weiss, on the other hand, provided a pre-developed system through which Obarzanek could develop his work. In this situation Obarzanek was reacting to the technology available to him, in much the same way as a composer of instrumental music reacts to the instruments available to represent his or her ideas.

In discussio with the author, a dancer who performed in both *Mortal Engine* and *the Quartet Project* found that working with Weiss's program was more predictable than Rothwell's, resulting in a performance in which they could focus on their aspect of the performance more easily.

In these the relationship between the technological output and the performers was to create a cohesive whole, in which all elements coalesce to form a single work containing each of the disparate elements.

2.1.2. Differentiating ACTIONA/V from previous approaches to integrating movement and technology.

The majority of performance situations that integrate movement, music and vision through computer technology follow similar paradigms to the integration of those elements without the use of computer technologies. In these cases the relationship between the various elements, lighting, sound and projections, are complementary at best, but very often given a supporting role to the choreography and subsequently the dancer(s) on stage, which is the paradigm that the dancer mentioned above, and which most other performers feel most comfortable with. In both *the Quartet Project* and *Mortal Engine* the role of the technician was in creating content that was complementary, or possibly the main focus of the work.

For much of the work of the practitioners considered above, their roles and contributions fit within this

traditional, well known, and useful paradigm. This particular paradigm is based on the very normal desire in the performance maker and their audience to place primacy on the human element of the performance. As humans we are naturally inclined to give our attention to that element within any communication process, and performance is no different.

ActionA/V differentiates from this approach in that its purpose is not to create work that unifies or coheres disparate elements. Instead it considers the process used by the action painters, one in which their movement was recorded and interpreted as a different artwork, to be viewed without and separate from the human element involved in its creation. Here, as the participants interact with the audio/visual interpretation of their movements, that interpretation is captured in the form of a movie, thus creating a work in a different format.

3. PROCESSES AND SYSTEM DESIGN

When developing computer based interactive systems, particularly those in which untrained participants are involved as the causative agent, it is important to create a balance between the interaction between the participant and the system being so predictable that the participants lose interest, and the converse, where participants lose interest because they do not experience a satisfactory sense of agency in the work.

When developing this software the original aim was to create an interactive system similar to that of a dancer improvising movement with another dancer, where there is an assumption that any interaction will be reciprocated in a recognizable way.

The intention was to develop a system that facilitates the translation of movement into sonic and visual elements. The system should create real time interaction between the movements of a participant and the audio/visual results of their movement that is recognizable, without being too prescriptive or overly predictable for the participant. The goal was to create a system that allows for a similar kind of interaction that one may have with another person, therefore having a fairly high degree of predictability, as well as a degree of novelty.

To facilitate this, an interactive environment using a stationary camera to gather data was created. For the original experiments and tests a studio space in the School of Creative Media, RMIT, that contained a video projector, screen and sound system was used. It was possible to darken the studio space sufficiently to illuminate only the performance area, which was, at the farthest point from the camera, approximately 3 meters wide by 2.5 meters high. The system was tested with four people who interacted with the system either as a pair or individually. When the participants were interacting with the image on screen their own image was not seen.

When moving in the performance area the actions of the participants created a change in an image on a screen. If there is no movement registered by the camera the screen showed black, but any movement triggered an image to be presented. These images were made from the primitive shapes, each of which could be represented as an outline, or solid three-dimensional object. These shapes could then be altered, from the participants' perspective, using processes described below.

The original software for this system was designed and implemented using Max/MSP/Jitter 4.6, in subsequent versions this was updated to Max5. The video camera used was an Apple iSight video camera connected to a MacBookPro via Firewire. Subsequent versions of ActionA/V do not require similarly controlled environments to those described above, but the less controlled the environment the less control the artist has in creating the audio-visual work as the possible interference from the environment will affect the outcomes.

3.1. Physical system

The video camera tracked movement in the performance space, and the resulting image was divided into twenty virtual quadrants. The aspects of the projected image that are influenced by movement in each of those quadrants are listed in Table 1, which also describes the attributes affected by movement in each of the twenty quadrants. The greater the amount of movement in a particular quadrant the more that aspect of the projected image was affected.

The proximity of the participant to the camera is important. As the participant moves further away from the tracking camera, the degree of movement in each quadrant must increase in order to have influence, and the possibility of movements being registered over more than one quadrant decreases. This means that the participant can exert more subtle control over the shapes and sound by their proximity to the camera. If the participant is closer to the camera, movements are registered over more quadrants, and less movement is required to influence the image, this offers an opportunity for very subtle and fine movements to be used. If the participant is further from the camera larger and grosser movements are required to generate an influence upon the projected image.

Figure 9 shows a dancer, in this case Brett Daffy, close to the camera and indicates the quadrants his position is affecting. Here the participant would be influencing: the image rotation on the z and y-axes, to a greater degree than the rotation on the x-axis, the lighting position and the camera angle x-axis as there is more action in those quadrants. Other aspects of the image, such as Camera angles 2 and 3, the Dimension value and the Blur amount, would not be influenced; the primitive image would also be changing.

Figure 10 shows the screen image corresponding to Daffy's position shown in Figure 9. This image is the combined result of many preceding motions of the participant, and therefore relates to many different influences, such as the participant having just moved rapidly through quadrants 3, 8 and 13, and is part of a continually evolving, dynamic process. The image as

seen by Daffy was in colour, comprising of blue, purple and green hues. Figure 11 shows movement in fewer quadrants than in Figure 9, due to Daffy being further from the camera.

3.2. Software System

As mentioned earlier, the software system used to create the system was the Max environment, and all objects are from those standard libraries. This allowed rapid development in an environment that allows a somewhat organic approach, with the capacity to experiment with different conceptual strands with an overall framework. The software has four basic areas, *Video capture*, *Sound generation*, *Video generation*, and *Output recording*. Each of these has sub-systems pertaining to their specific functions.

3.2.1. Video capture

Video capture converts the image of the performance space and all activity within it to a grayscale image in real-time, splits that image into 20 equally sized quadrants, and each of these affect different aspects of the image created in Video generation (see Table 1).

A temporal differentiating process measures the activity in each quadrant. Figure 1 shows these steps in a sub-system of *Video capture*. A similar sub-system was used for recognizing motion in each of the 20 quadrants.

Figure 2 shows the matrix used to parse the information from the Motion recognition system and then supply it to the *Video generation* system. Here the first five "p" objects, marked row1, influence the rotation of the image on the x axis, the position of the light source on the x axis, how the image is perturbed, the position of the light source on the y axis, and the rotation of the image on the y axis, row2, row3 and row4 follow in a similar way, with each of the "p" objects relating to the aspect of the image outlined within the object.

The information regarding degrees of motion is fed to the related aspects of *Video generation* from each of these sub-systems.

3.2.2. Video generation

The process of creating the videos was based on altering seven primitive shapes from within OpenGL 3D graphics: sphere, torus, cylinder, opencylinder, cube, opencube, plane, and circle. These shapes are selected by the participants' action as they moved in the quadrant relating to that aspect of the image. In earlier versions of the program the number of dimensions on the *x* plane could be dynamically altered through the action of the participant, as can the size of the inner circle of the torus shape, should that be the selected shape. This section of the video generation process is shown in Figure 3.

These shapes were then altered through the perturbation/distortion process outlined in Figure 4. Here the mathematical functions built into the Max program are used to dynamically alter the original, primitive, shape through the use of the basis function graphing process built into the Max environment. This allowed quite significant changes to the shape in accordance with the degree of activity in the quadrants relating to that shape.

The type of basis function used in the experiments was the transfer.sine function, as this requires less processing power than other basis functions, and resulted in more intuitively predictable outcomes for the participants. The basis scale, basis function time offset, and the basis function displacement operate together to influence the degree of distortion of the original shape.

In later versions of the program the movements of the participant could select theses shapes, however in the version used in the experiments at RMIT, the shapes used were sphere and torus.

The system for determining the virtual positions of the participants when viewing the virtual objects is shown in Figure 5.

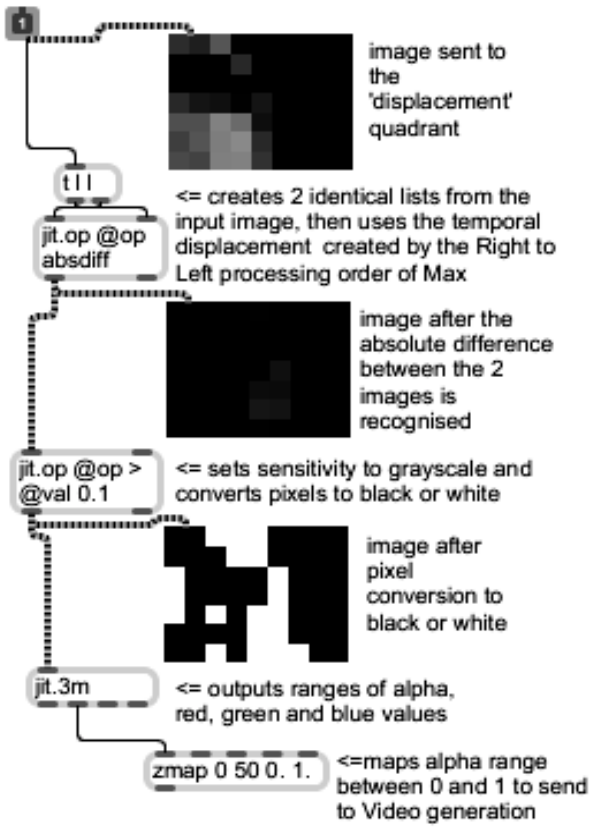


Figure 1: Motion recognition system

Using these three attributes to orient the image from the viewers' perspective, in effect, simply changes the image as seen on screen. However, what is does facilitate is a very subtle control of the perceived shape of the image, as miniscule changes to particular aspects of the image can be wrought through very subtle movement by the participant.

The image is then rendered using the drawing modes, as shown in Figure 6. The possible modes are: points, lines, line_strip, line_loop, triangles, tri_strip, tri_fan, quads, quad_strip, polygon, or tri_grid. As these different modes are applied, the sense of the shape alters dramatically, as for example, the spaces inside an outline are filled, or just the outline is shown. It is possible that the movements of the participant can alter these aspects dynamically, depending on the drawing mode selected, resulting in a mixture of filled in or un-filled in shapes being seen.

Finally colour is applied to the image, also shown in Figure 6, and this is done through the participant moving through the quadrants relating to the red, green and blue hues. The sense of shading, intensity, and colour is influenced by the blend mode selected, which creates more interesting effects than simply using the alpha channel to create transparency when rendering colours to the objects generated.

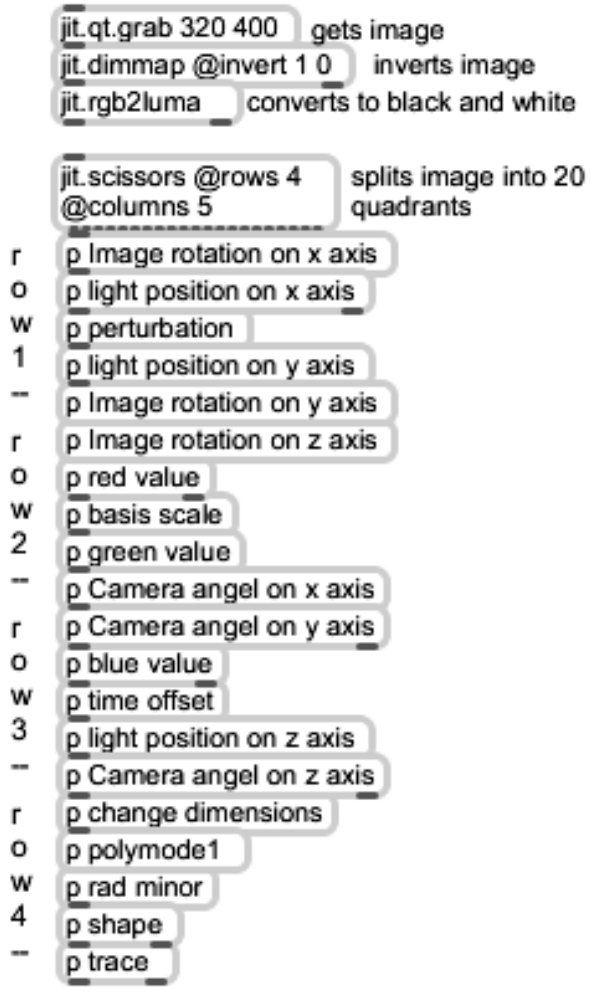


Figure 2: Motion capture parsing matrix

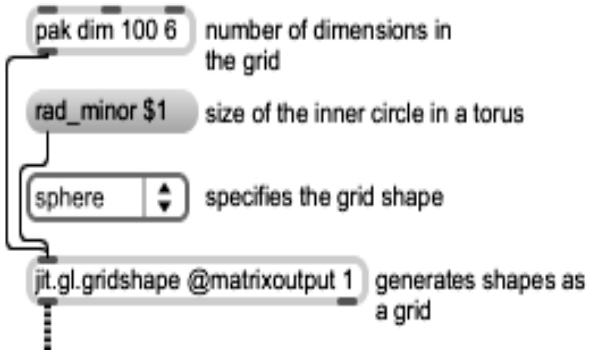


Figure 3: Process for creating simple shapes

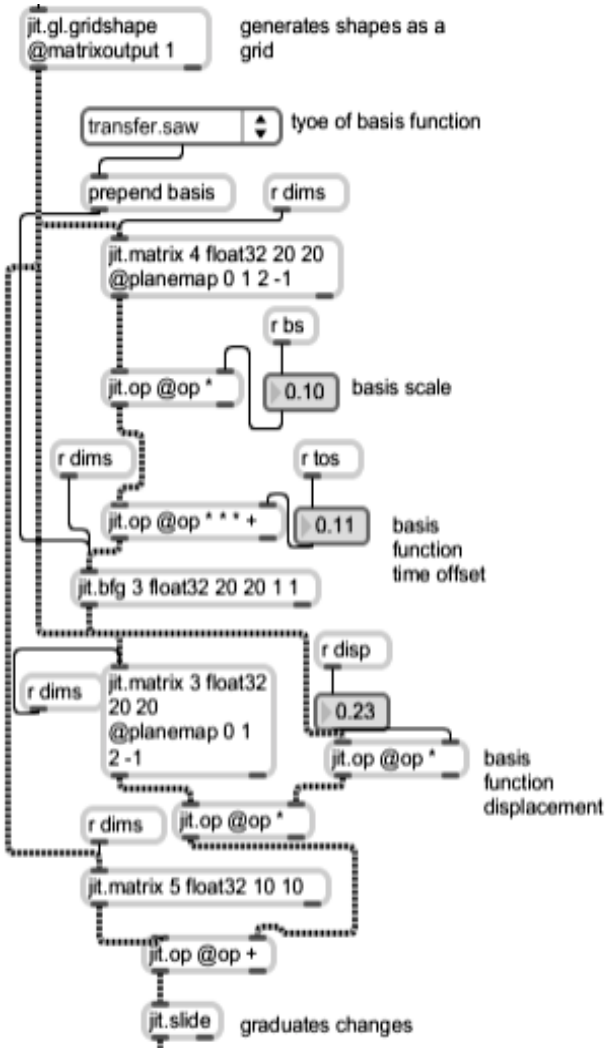


Figure 4: Process for perturbing the simple shape

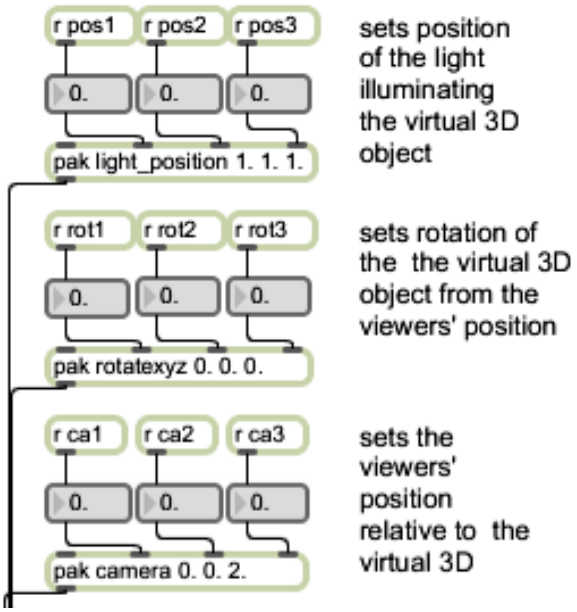


Figure 5: Process for setting the lighting position, rotation, and viewers' position relating to the virtual 3D object

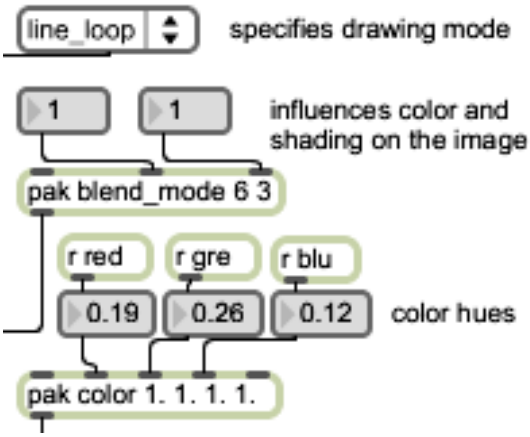


Figure 6: Process for adding colour to the virtual object.

3.2.3. Sound playback and dispersion

In the RMIT experiments, sound playback was done using five separate sample playback units that play continuously after being triggered at evenly spaced time intervals, shown in Figure 7 below. The result was a polyphonic texture built on a displacing, looping structure in which each part interacts with the other four to create a five-part canon; the sounds heard when the participants were interacting with the process are described in section 4, below. This system is also used in later versions of *ActionA/V*.

These five sound sources are routed through a matrix that gates the sound signal and specifies which sounds will be played through a stereo output. Each sample player is allocated to a specific input on the matrix which can then be output either of the stereo outputs. The output of each sample player is selected through the movements in ten of the quadrants around the periphery of the performance area.

While the samples are continuously looping, the motion detected in the performance space, and thus the mover in the space, decides which and when the sounds are heard via the gating/matrix process by selecting the sounds that pass through the gate. This allows the possibility of melodic and polyphonic outcomes, with each of the five possible sounds being triggered by movement in the performance space.

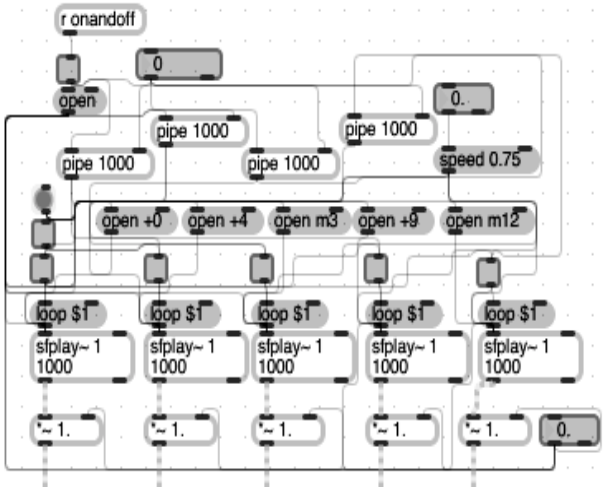


Figure 7: Sound file playback

The pitch intervals use in the RMIT experiment comprised a major pentatonic scale, with the tonic repeated octave higher. This choice of intervals allowed an outcome that is inherently melodically and polyphonically pleasing to the ear, within the typical Western musical traditions.

While the global inter-onset time between sounds was set, as were the durations of each sound, the actual effect was the creation of a participant controlled melody/polyphony that influenced the improvisation that generated the works.

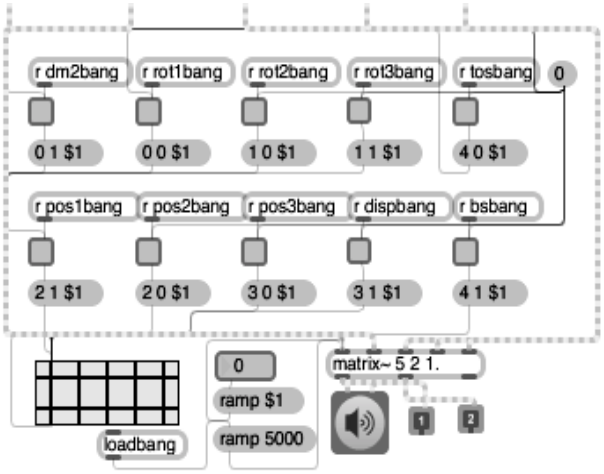


Figure 8: Sound dispersion process

3.2.4. Output recording.

The video and audio output generated by the participants at RMIT was recorded as .jpeg files. A video of the performers as they interacted with the program was also taken. This allowed the reinterpretation of their actions to create other works built from the program that they interacted with, or from developments of that program.

4. PARTICIPATION IN THE RMIT TESTS

Three dancers of various experience, focus and skill, and one actor, whose practice has a movement focus, participated in the experiment and development of the work(s). This occurred in a studio over four, two to three hour, sessions. The were not given any direction, but could move within the performance space and could orient themselves to the motion-tracking camera and the screen that showed the results of their movements in real time as they wished.

These dancers were asked to improvise with the image seen on screen and the sound that they heard in much the same way that they would interact with another dancer in an improvisational setting, considering their interaction as they would if improvising with another dancer. They were then asked to interact with the sound and moving image in order to create images and sound that were pleasing to them, or that they feel have some artistic validity or merit. Therefore they were asked to create a work in a form other than dance through their dance-like and dance-informed actions.

When introduced to the project participants were given a very rough outline of the project, being told simply that their movements would be interpreted as sounds and abstract images on a screen. This ensured that the participants had a clear idea of their role and the potential outcomes of their interaction with the program when they arrived for the experiments. During the process the dancers could not see their bodies on screen, only the results of their actions. This was so that they were not overly distracted by images of themselves, instead giving their attention to what they saw as a result of their movements more than themselves.

The process for developing an understanding of how to interact was essentially heuristic, following the trajectory: discovering what the images would do when they moved, how they could control the images, and how they could interact with the images as if they were another agent. After about twenty minutes of improvising the participants felt that they could collaborate with the images, as if with partner in developing an improvised work, after which they began to see their role as creating an audio-visual work and being comfortable with that role and agency.

When the participants were improvising with the screen they would often be surprised at how long they had been interacting with the screen; in one case a participant assumed that they had been interacting for about 10 minutes when in fact they had been interacting for about 40 minutes.

4.1. Creative descisions in the original tests

The sound for the original tests was generated from two sources, a sound file comprised of a solo voice sourced from the Logic 8 (© Apple 2007) sound library titled *Religious Voice*, which was convolved with a recording the speech Ronald Reagan gave discussing the Challenger Space shuttle tragedy of 1986 [11] using SoundHack (© Tom Erbe). This resulted in a sound-scape that had reference to the human voice in both song and speech, but without direct denotative aspects of song or speech.

As mentioned earlier, it is possible to create a melody by triggering the different sounds through their movement in the performance space, thus creating a melodic outcome through their actions, just as they create a visual outcome through their actions.

The rhythmic structures within each of these two sounds provide an overall rhythmic counterpoint, the religious voice being melismatic, and Reagan's speech being clipped and segmented, but still maintaining a sonorous sound. In both cases Reagan and the religious singer made extensive use of intonational devices to influence emotional responses in the listener.

The quadrant-based paradigm is used as it is seen as making an interface similar to that of a musical instrument. A piano, for example, has up to 88 quadrants, the keys, and each causes a specific sound to occur when physically activated. All musical instruments provide similar interfaces that translate a physical gesture into a sonic gesture; some allow a greater degree of interaction and control between the physical action and the sonic result. Here it is possible for the participants to cause responses in any or many of the quadrants, much like a pianist can depress one key or many keys simultaneously by moving closer or further away from the tracking camera, and to actively influence the output of each quadrant by the degree of movement made in that quadrant.

It was important that participants be able to hear and see the program reacting to their movements in a predictable way, similar to the way that a piano reacts predictably. Conversely it was important that the participants were not bored by the results of their interaction by those results being too predictable.

While the participants were interacting with the system they did not see their own image, this meant that they were not distracted by any considerations of their own movement other that how it affected the screen image. This resulted in subtler and more refined movements that focused on developing the onscreen image and that reduced their focus on the physicality of their movement as their familiarity with the results of their actions developed.

When deciding what influences to be attached to each quadrant the shape of a person standing in the centre of performance space was considered, as seen in Figure 9. Here movement in the upper central column of quadrants influences the changes made to the image

shape but not original primitive shape, the quadrants both sides of that central column influence the colour and virtual lighting position, and the external columns influence the position of the virtual viewer. The bottom row influences the shape of the primitive and how transitions in that shape occur, this means that primitive remains constant unless there is a deliberate desire by the mover to change it. The results of the movements seen in this image would include much change in the shape of the primitive and the orientation of the viewer to the shape, but little activity in the colours seen on screen.

It was important that when interacting with the system the participants saw their movements represented in an unfamiliar way, so that they did not revert to familiar styles or clichés similar to those they may use when interacting with another dancer as they were interacting with the screen image, and so that they saw their movements being represented in a very different medium.

4.1.1. Image decisions

The images generated through the actions of the participants bear no resemblance to their actions other than to reflect the gestures as documented in the movies made. This has a similarity to the process of the action painters, with their actions being recorded in the still images that they created.

It was also important to ensure that there was as little as possible self-consciousness in the participants in the development of the *ActionA/V* documents. When dancers are developing their skills they are required to develop an awareness of their position in space, and in relation to other dancers around them, and to develop a sense of the quality of their movements within the various styles of dance. This consciousness often pervades their intuitive movements regardless of immediate context and purpose and can result in the unintentional use of clichés.

By removing the image of themselves from their immediate interaction, it was thought that the participants would possibly reduce their automatic use of and reliance on the tropes, conventions and clichés often apparent as a result of extensive training. It was also important that the participants consider the output of their actions as separate from themselves, much as the action painters considered their work on canvas as separate from their physical actions when developing the work. This was usually the case, with the participants quickly developing new libraries and styles of movement different from those they would normally use in an improvised movement setting, as they interacted with the sound and images generated from their movements

4.1.2. Audio decisions

The reason for convolving the two audio tracks in the initial tests discussed in 4.1 was to create an audio experience that could be thought of as complementary to the visual aspects of the work when the participants interacted with *ActionA/V*.

It was important that the participants sense a direct relationship and coherence between the sound and image. As the images on screen were flowing in their movement, unless the participant moved very suddenly and vigorously, it was decided that the sounds used should in some way reflect this. The movements of the participants influenced the playback of the sounds, as more active movement created more active melodic and polyphonic outcomes.

5. CONCLUSION

ActionA/V has developed from its initial driving concept, that of creating a system through which the processes of developing an artwork through the tools of one mode creates an artwork in a different mode, to be an interactive system that is both entertaining and interesting to the casual user and fulfills that initial idea.

Conversations with the participants in the RMIT experiments indicated that they were able to quickly understand the relationship between their actions and the movements seen on screen and the sounds heard, and to then develop a novel movement vocabulary that related directly to what they perceived as the vocabulary of the sound and images generated form their movements.

They also indicated that being involved in the project did cause them to reconsider their approach to movement and dance in an improvised and in more predetermined settings, in particular enjoying the ability to use very subtle movements to cause either subtle or gross outcomes, and vice versa.

Regarding the development of a system for translating processes across art forms, *ActionA/V* allows direct interaction between modes of expression. Here kinetic movement is seen to be creating a response, in a different medium, which has a direct relationship to that movement but is not in itself kinetic.

This method of translation allows for many other processes to be used in developing future works; any kind of motion or visual input can be used to generate the audiovisual material, and it is to be used for experiments and generating future works.

6. REFERENCES

[1] Barrios, M. (2008, November 9). "Nick Rothwell: Augmenting Choreographers Minds." from http://www.dance-tech.net/video/nick-rothwell-augmenting.

[2] Birringer, J. (2010). "AlienNation people." Retrieved April 7, 2010, from http://www.aliennationcompany.com/people/jobi.htm.

[3] Company in Space. "http://www.companyinspace.com/front/cis_fs.htm." Retrieved April 5, 2010, 2010.

[4] Dean, R. (2010). "austraLYSIS." Retrieved April 6, 2010, from http://www.australysis.com/.
[5] Edwards, G. (2010). "Dance Review: Mortal Engine." trespass magazine, from http://www.trespassmag.com/review-mortal-engine/.
[6] Karmel, P. (1998). "Pollock: Struggle and Mastery." MoMA, **1**(6): 2-5.
[7] Medlin, M. (2010). "Critical Path." Retrieved April 6, 2010, from http://www.criticalpath.org.au/.
[8] Namuth, H. (1951). Jackson Pollock. United States, Hans Namuth Ltd.
[9] Paine, G. (2010). "activatedspace." Retrieved April 5, 2010, 2010, from http://www.activatedspace.com.
[10] Polcari, S. (1990). "Martha Graham and Abstract Expressionism." Smithsonian Studies in American Art **4**(1): 3-27.
[11] Reagan, R. (1986). American Rhetoric: Top 100 Speeches, AmericanRhetoric.com.
[12] Rosenberg, H. (1961). "The American Action Painters." The London Magazine **1**(4).
[13] Rothwell, N. (2009). "Quartet Project." Retrieved April 11, 2010, from http://www.quartetproject.net/space/start.
[14] Rothwell, N. (2010). " cassielmusic, media, systems, performance, installation." Retrieved April 7, 2010, from http://www.cassiel.com/space/.
[16] Rubidge, S. (2010). "sensedigital." Retrieved April 7, 2010, from http://www.sensedigital.co.uk/.
[17] Sky, H. (2010). "Hellen Sky." Retrieved April 6, 2010, from http://www.hellensky.com/hellen.swf.
[18] Sky, H. and G. Paine. (2010). "activatedspace." Retrieved April 5, 2010, 2010, from http://www.activatedspace.com.
[19] Troika Ranch. (2010). "Troika Ranch." Retrieved April 4, 2010, from http://www.troikaranch.org/.
[20] Unknown. (2010, 2010). "dance technology." Retrieved June 2, 1020, from http://www.search.com/reference/Dance_technology.
[21] Wanderley, M. (2004, April 22 2004). "Interactive Systems and Instrument Design in Music (Working Group)." Retrieved April 4, 2010, 2010, from http://www.music.mcgill.ca/musictech/ISIDM/wg.html.
[22] Weiss, F. (2010). "This is the web site of Freider Weiss." Retrieved April 11, 2010, from http://www.frieder-weiss.de.

1. Rotation on the *x* axis of the original image	2. Light position 1 relative to the position of the original image	3. Perturbation 1. Influencing the shape of the object	4. Light position 2 relative to the position of the original image	5. Rotation 2 on the *y* axis of the original image
6. Rotation 3 on the *z* axis of the original image	7. Red value	8. Perturbation 2. Influencing the shape of the object	9. Green value	10. Camera Angle 1 on the *x* axis relative to the position of the original image
11. Camera Angle 2 on the *x* axis relative to the position of the original image	12. Blue value	13. Perturbation 3. Influencing the shape of the object	14. Light position 3 relative to the position of the original image	15. Camera Angle 3 on the *z* axis relative to the position of the original image
16. Number of dimensions of the original image	17. Image type (alternating between outline or solid)	18. Radius minor (the size of the hole in the centre of the torus)	19. Shape primitive (sphere, cylinder, cube, or torus)	20. Blur amount, influencing the roughness of the image edges.

Table 1: Aspects of the video influenced by motion in the 20 quadrants

1. Rotation on the *x* axis	2. Light position 1	3. Perturbation 1	4. Light position 2	5. Rotation 2 on the *y* axis
6. Rotation 3 on the *z* axis	7. Red value	8. Perturbation 2	9. Green value	10. Camera Angle 1 on the *x* axis
11. Camera Angle 2 on the *x* axis	12. Blue value	13. Perturbation 3	14. Light position 3	15. Camera Angle 3 on the *z* axis
16. Dimension value	17. Image type (alternating between outline or solid)	18. Radius minor (the size of the hole in the centre of the torus)	19. Shape primitive (sphere, cylinder, cube, or torus)	20. Blur amount, influencing the roughness of the image edges.

Figure 9: Movement in many quadrants



Figure 10: Resulting image from the movement in the quadrants shown above

1. Rotation on the *x* axis	2. Light position 1	3. Perturbation 1	4. Light position 2	5. Rotation 2 on the *y* axis
6. Rotation 3 on the *z* axis	7. Red value	8. Perturbation 2	9. Green value	10. Camera Angle 1 on the *x* axis
11. Camera Angle 2 on the *x* axis	12. Blue value	13. Perturbation 3	14. Light position 3	15. Camera Angle 3 on the *z* axis
16. Dimension value	17. Image type (alternating between outline or solid)	18. Radius minor (the size of the hole in the centre of the torus)	19. Shape primitive (sphere, cylinder, cube, or torus)	20. Blur amount, influencing the roughness of the image edges.

Figure 11: Movement in few quadrants due to distance from the camera